Chaos Engineering
: Failing on Purpose

2026-02-26, 김찬빈 (DevRel)

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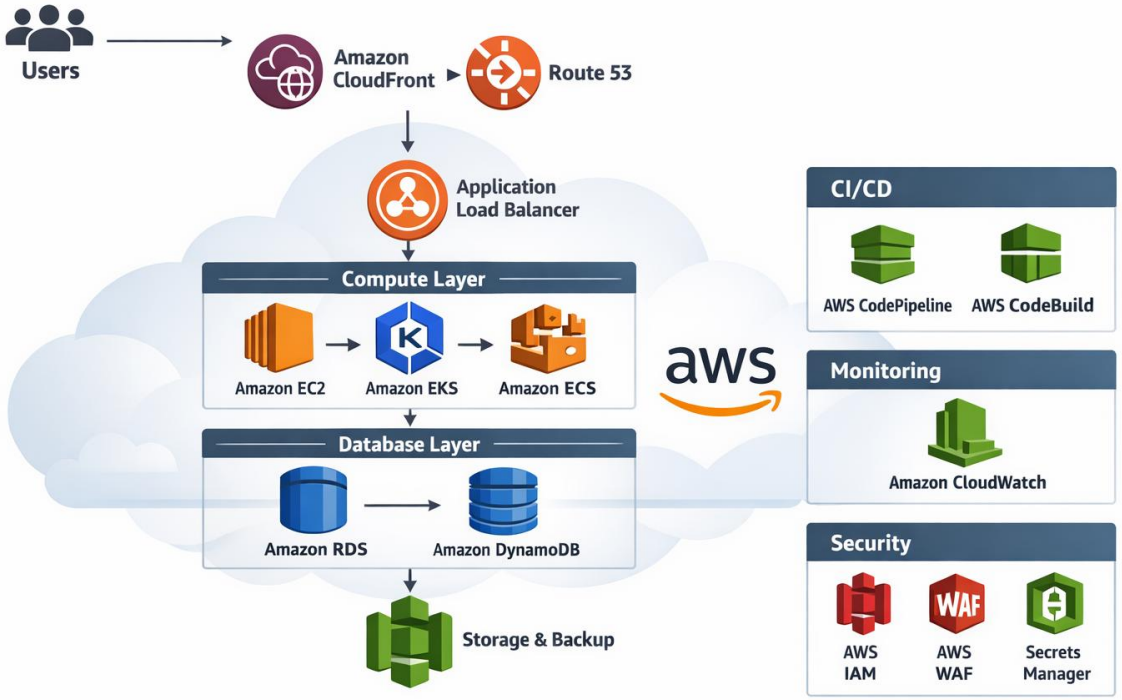

System Engineering,

But – Are We Ready to Break?

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We Think We Are Ready

아마도, 우린…

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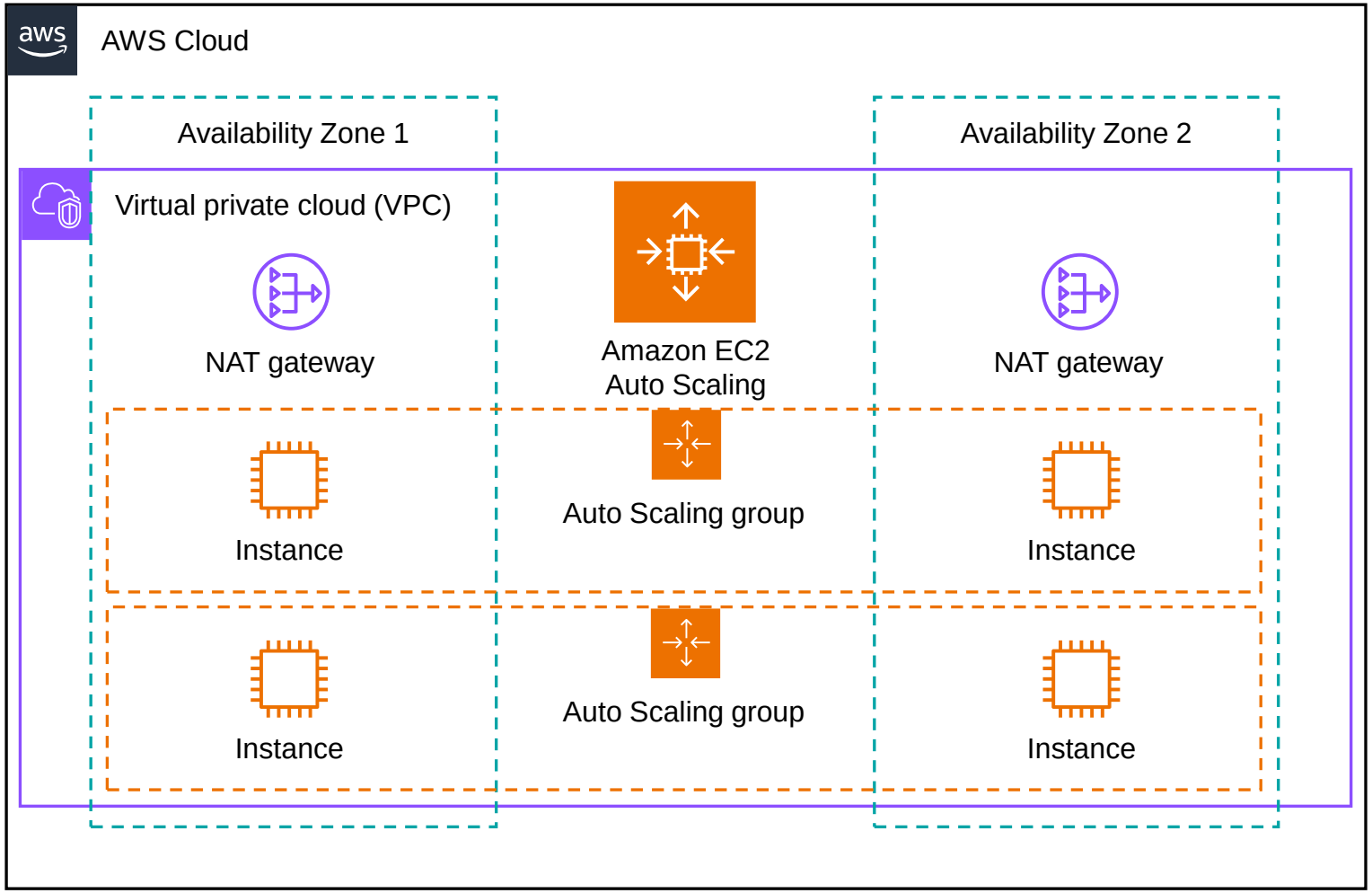
We Think We Are Ready



Jenkins

Grafana

argo

But…

Have We Tested Failure?

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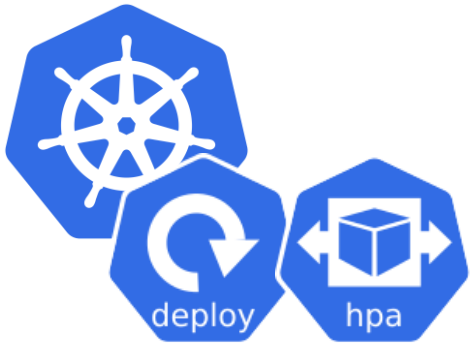
We Think We Are Ready



- No Downtime?
- Multi AZ?
- Auto Scaling?

Or.. Survive Failure?

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We Think We Are Ready

Hardware Fails.

Network Fails.

Software Fails.

Humans Fail.

실패는 특별한 사건이 아니라, 전제 조건임

Ex) 자동차 브레이크

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So, We Experiment.

What is Chaos Engineering?

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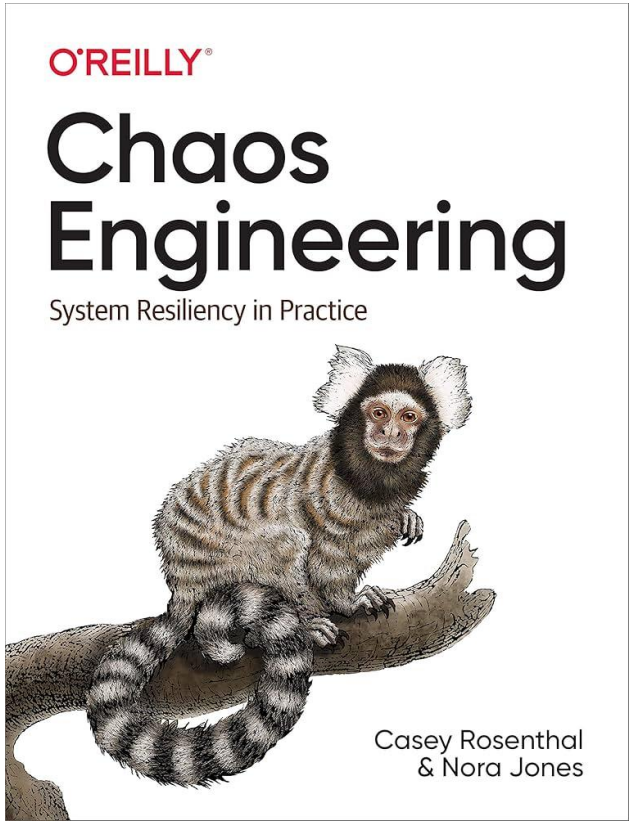
So, We Experiment.

1. 정상 상태 기준 수립
2. 가설 수립
3. 실험
4. 결과 분석



Chaos Engineering

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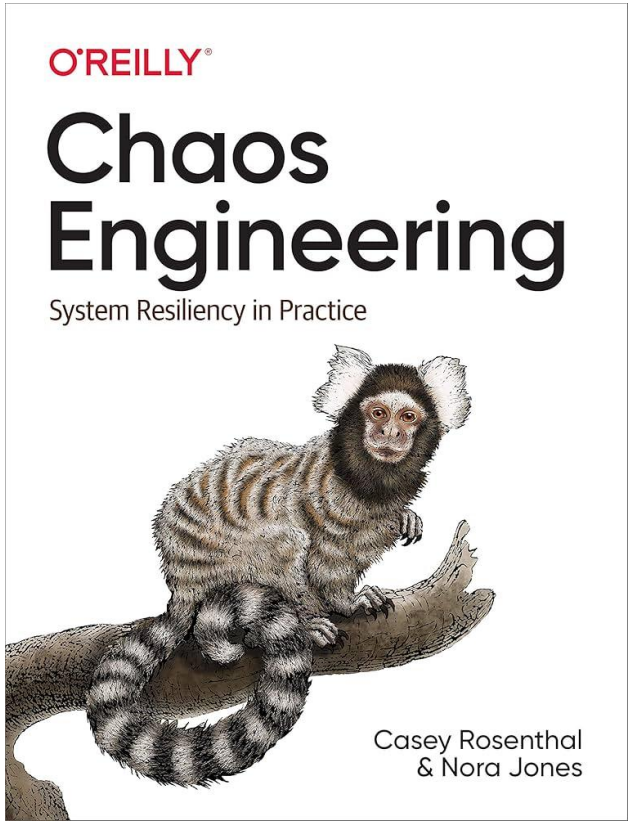
So, We Experiment.



Chaos Engineering

실패를 의도적으로 주입하여

시스템의 신뢰성을 검증하는 방법

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So, We Experiment.

Chaos Monkey

Simian Army

- Chaos Monkey
- Chaos Gorilla
- Chaos Kong
- Latency Monkey
- …

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So, We Experiment.

Understanding the AWS October 2025 Outage: What Really Happened Beyond "Just Use Multi-Region"

Rohit Sharma
Associate Director – AI | Principal AI Architect | Agentic AI, GenAI, Enterprise AI

2025년 10월 22일

Like most of us - I saw the news of AWS outage affecting 1000s of servi
globally. My first reaction was same as the industry's:

> Weren't these organizations using multi-region architec

https://www.linkedin.com/pulse/understanding-aws-october-2025-outage-what-really-hap

■ **이용자 여러분께 드리는 말씀**

이번 장애는 시그널링이 사용하는 인프라(AWS)의 특정 리전에서 발생한 글로벌 이슈로, 저희가 사전에 예측하거나 제어하기 어려운 **불가항력적인 장애**이지만, 그럼에도 불구하고 서비스 이용에 불편을 드린 점 진심으로 사과드립니다.

클라우드

AWS 미국 동부 리전 대규모 장애 발생...전 세계 주요 서비스 마비

발행일 2025년 10월 20일 **읽는 시간** 2분 30초

> *AWS 미국 동부 리전에서 발생한 서버 장애로 코인베이스, 퍼플렉시티, 포트나이트 등 전 세계 주요 온라인 서비스가 동시다발적으로 마비됐다.*

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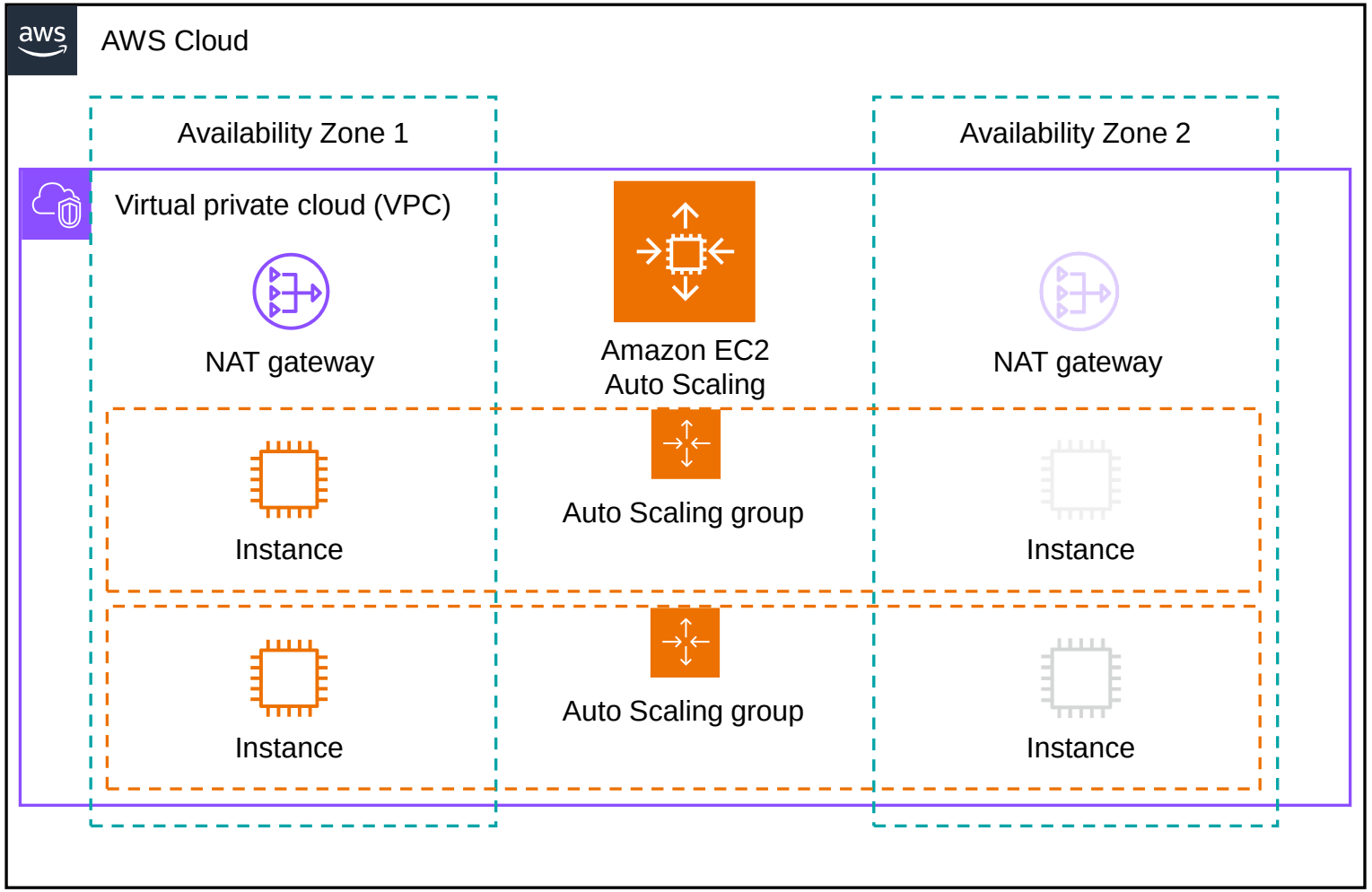
So, We Experiment.



실제 Product 환경에서 수행

비행기를 날려 놓고,
엔진을 하나씩 꺼 보는 테스트

-> 즉시 중단 가능하도록 조치…

-> 예측 가능한 실패를 검증하는 것

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So, We Experiment.

AWS
Fault Injection Service

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Experiment

직접 망가뜨려보겠습니다.

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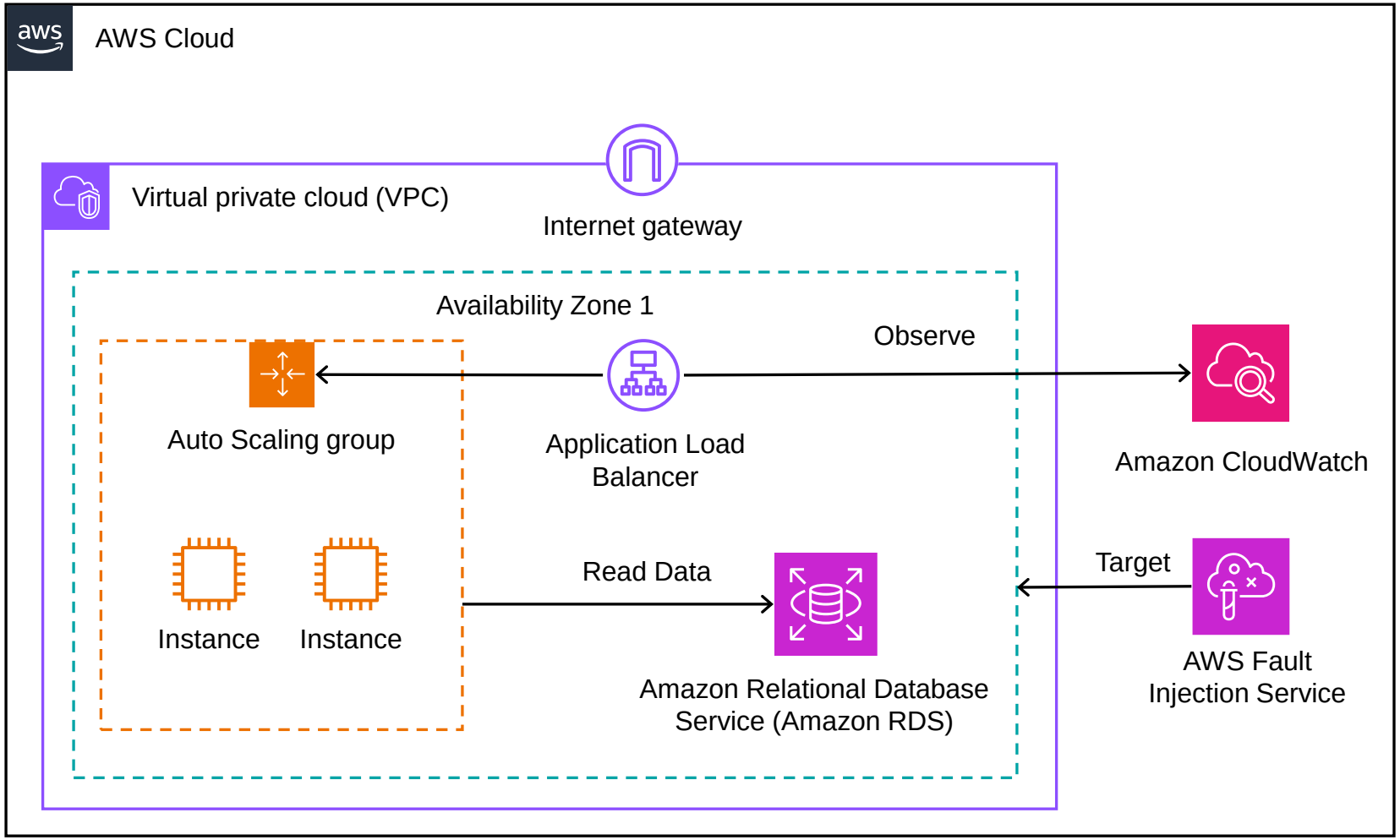
Experiment



과연 문제가 있을까?

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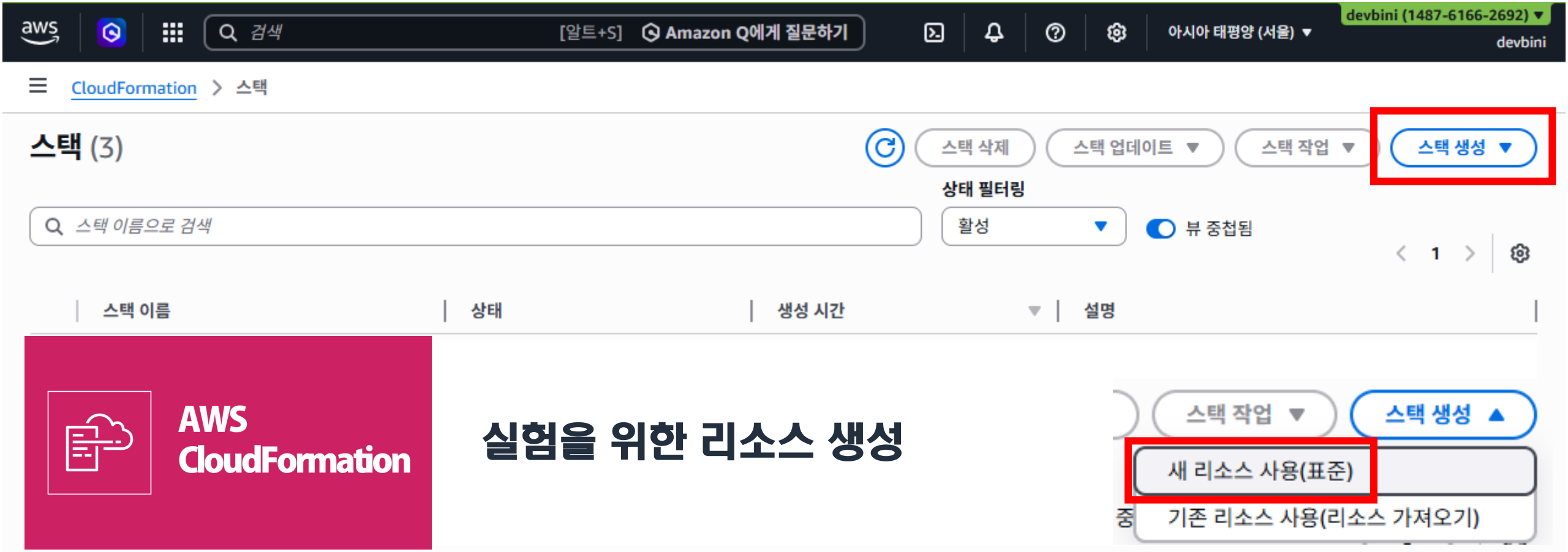
Experiment



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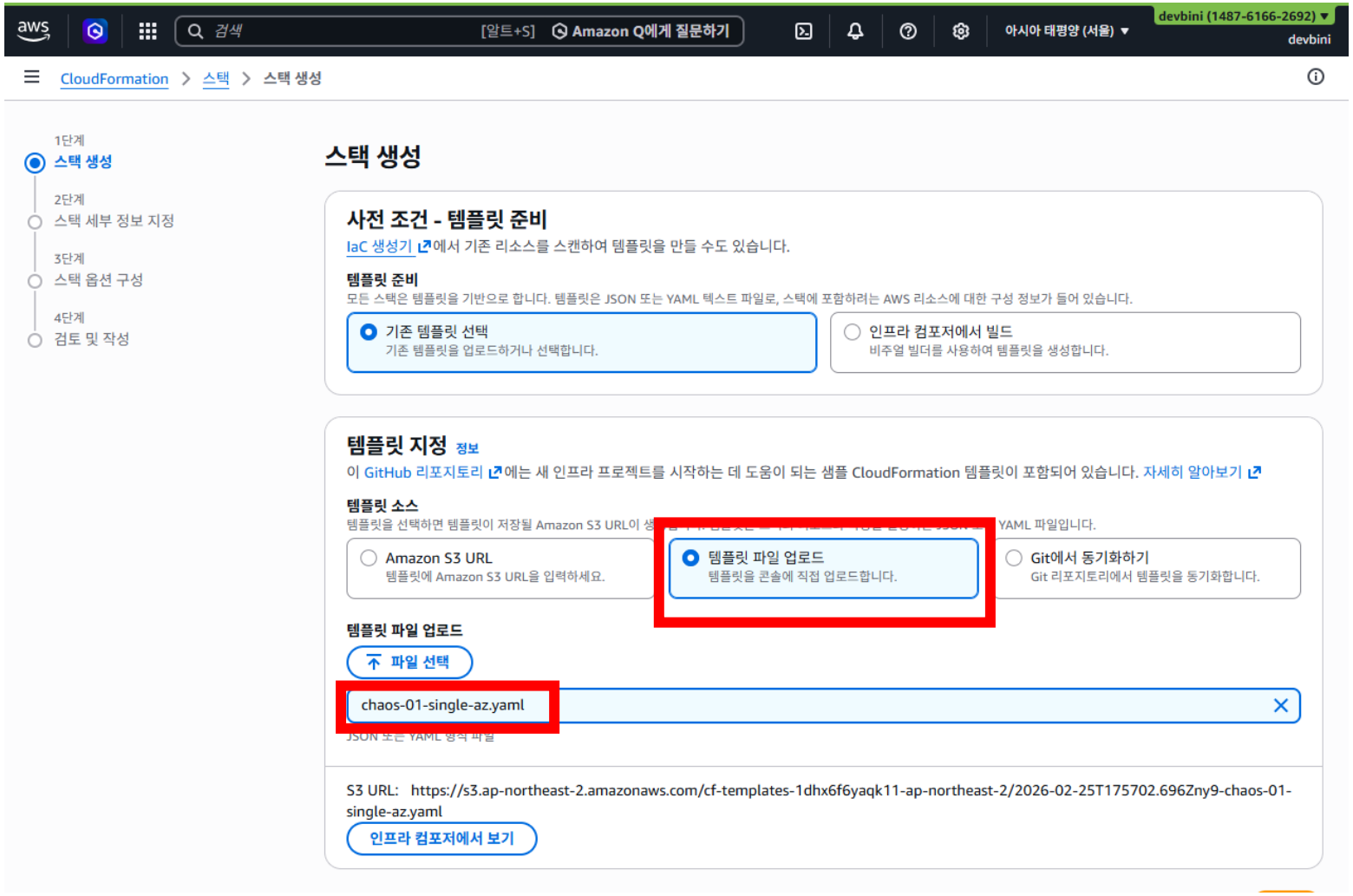
Experiment



기존에 제공 된 .yaml 템플릿 기반 리소스 생성

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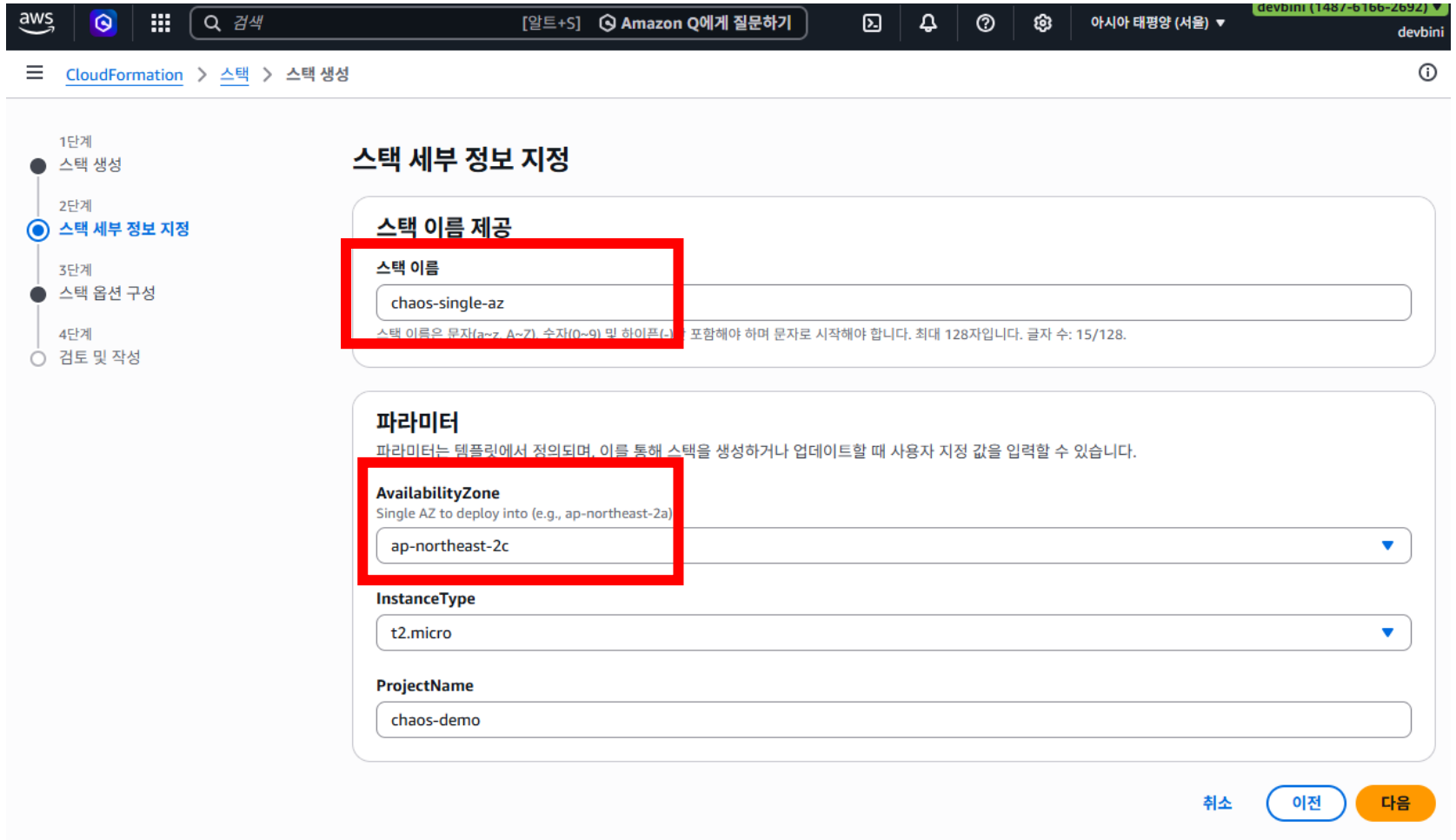
Experiment



기존에 제공 된 .yaml 템플릿 기반 리소스 생성

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Experiment

기존에 제공 된 .yaml 템플릿 기반 리소스 생성

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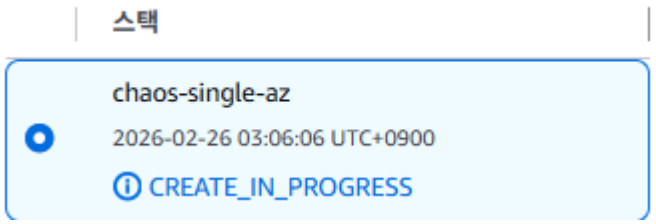
Experiment

기존에 제공 된 .yaml 템플릿 기반 리소스 생성



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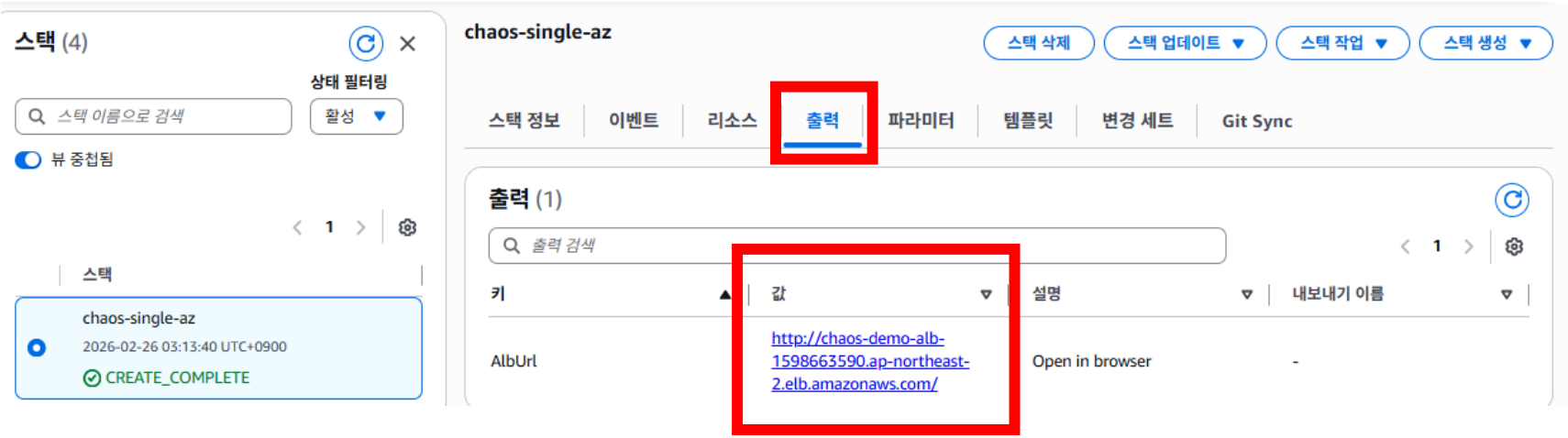
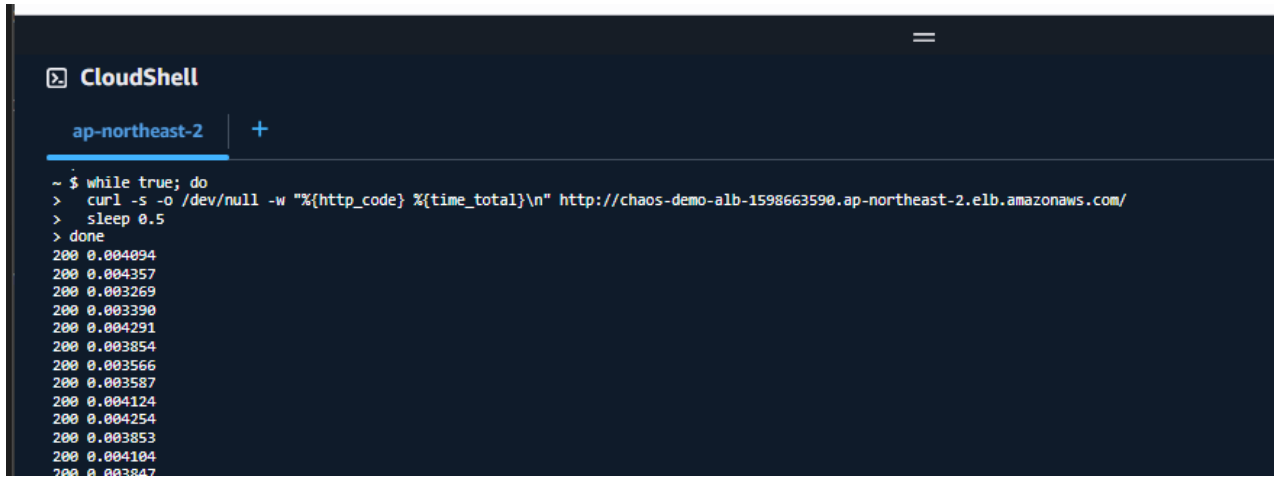
Experiment





생성 된 AlbUrl 사용

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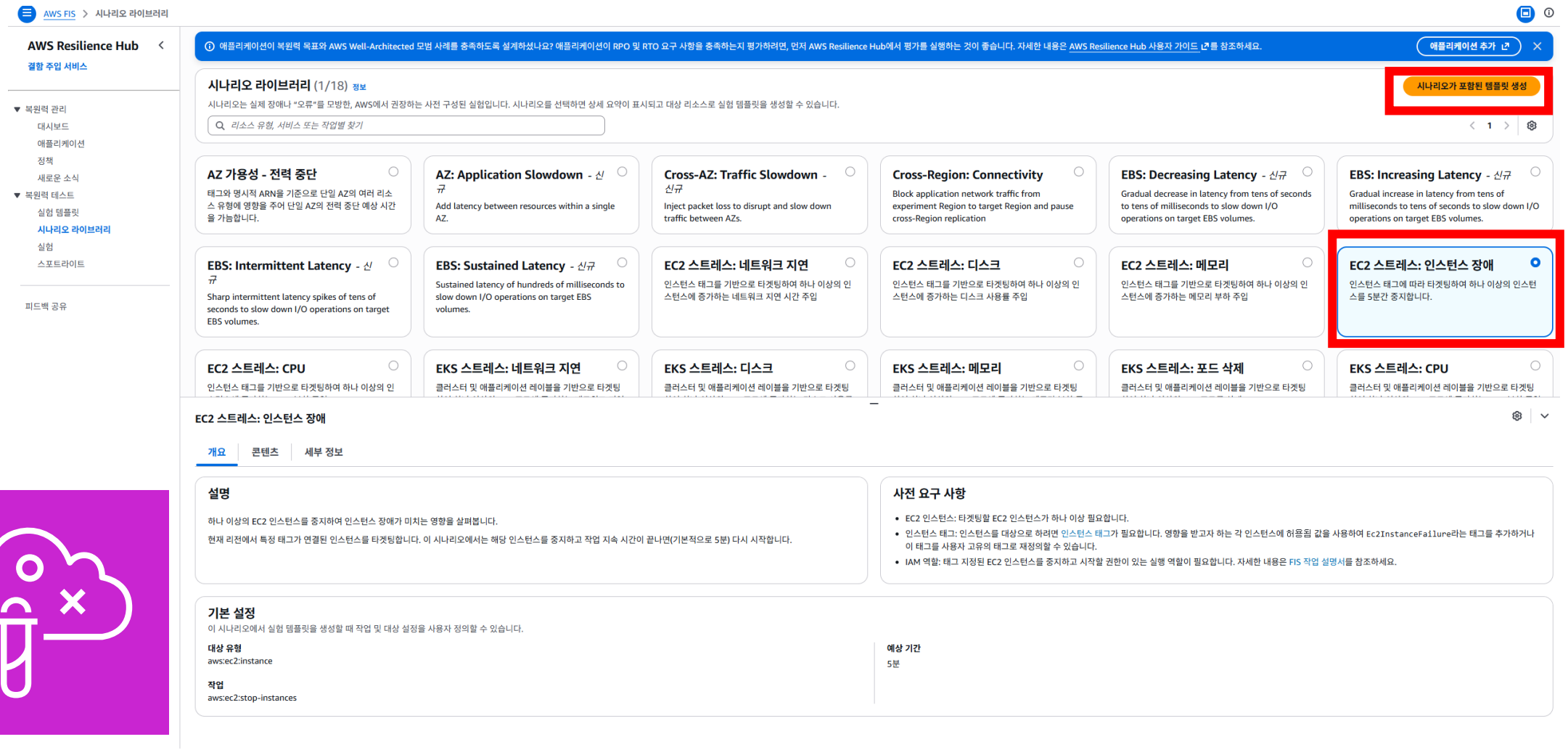
Experiment



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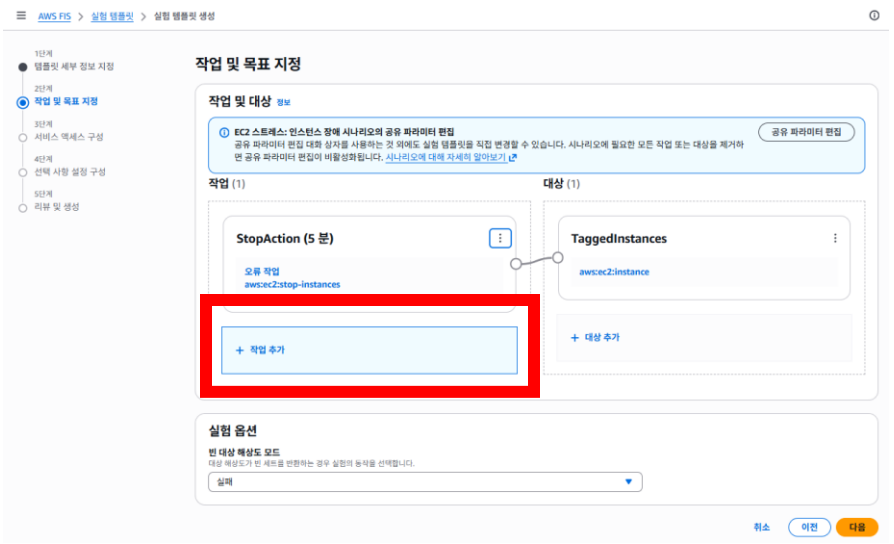
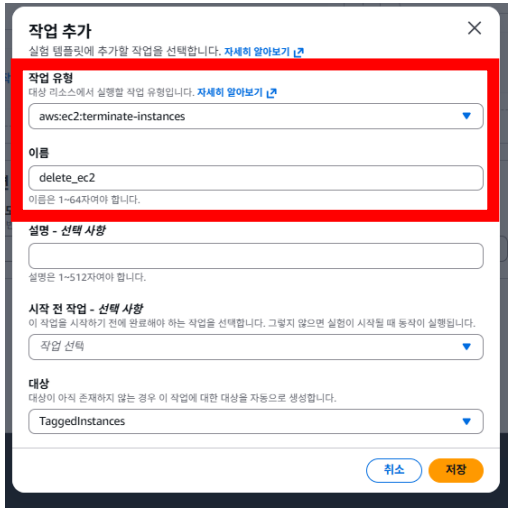
Experiment



인스턴스를 삭제하여 테스트할 것.



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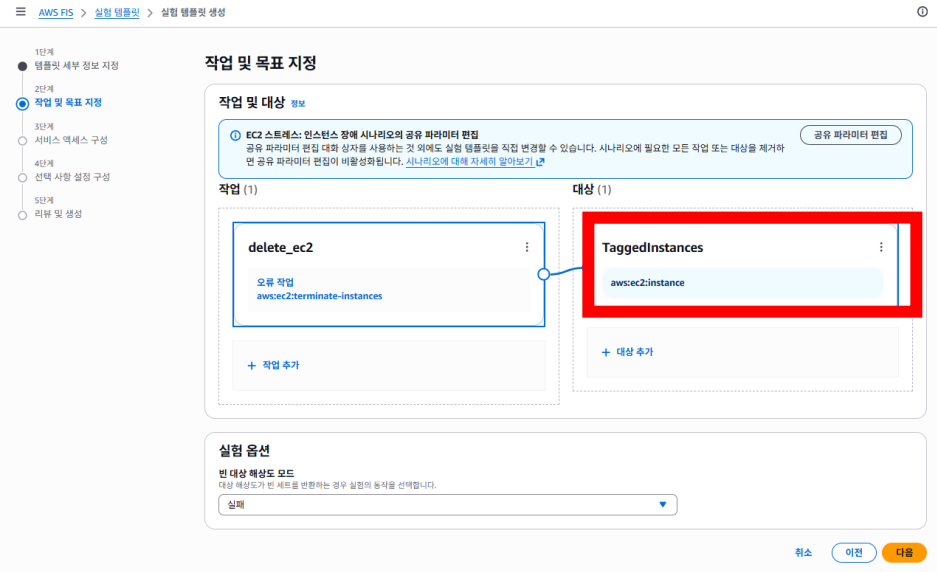
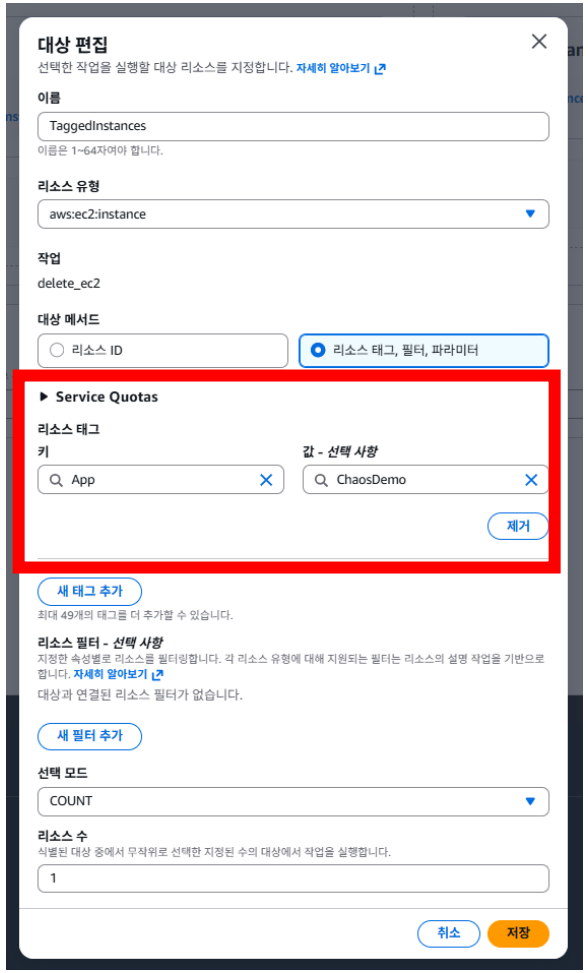
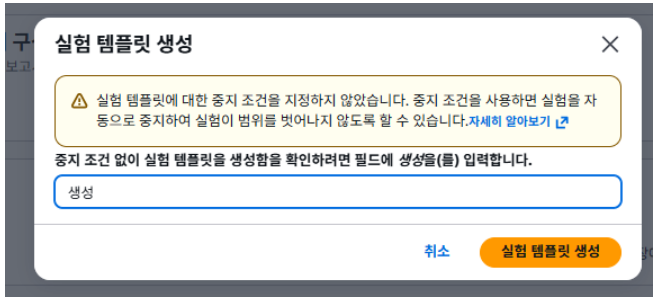
Experiment







리소스 필터 변경
(Cloud Formation으로 생성함)

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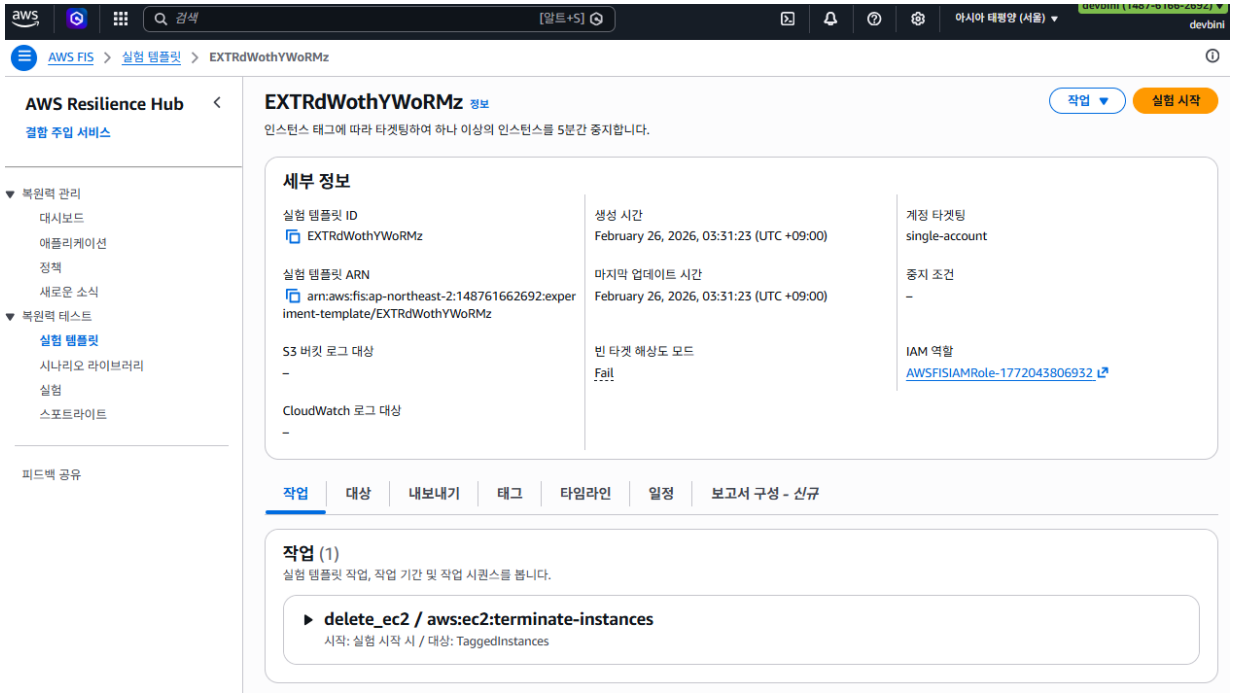
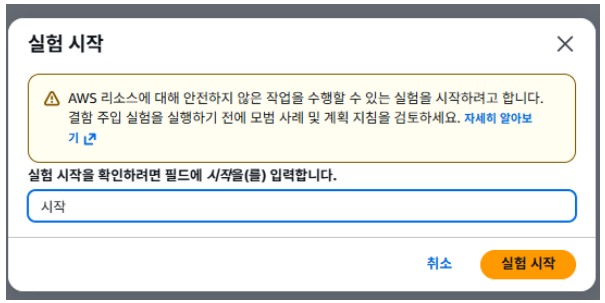
Experiment





실험 시작 시,

실제로 해당 시나리오가 시작됨.

구동 시간만큼 금액이 발생함.

(장애는.. 더 비쌉니다.)

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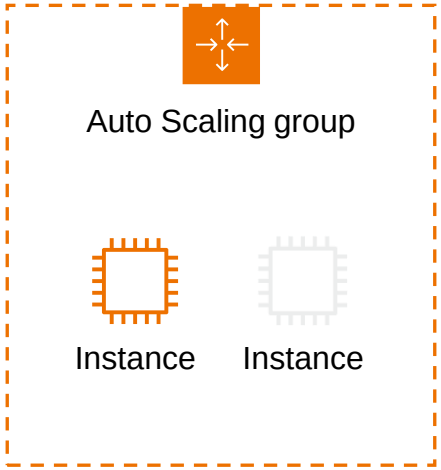
Experiment



어떻게 될까?

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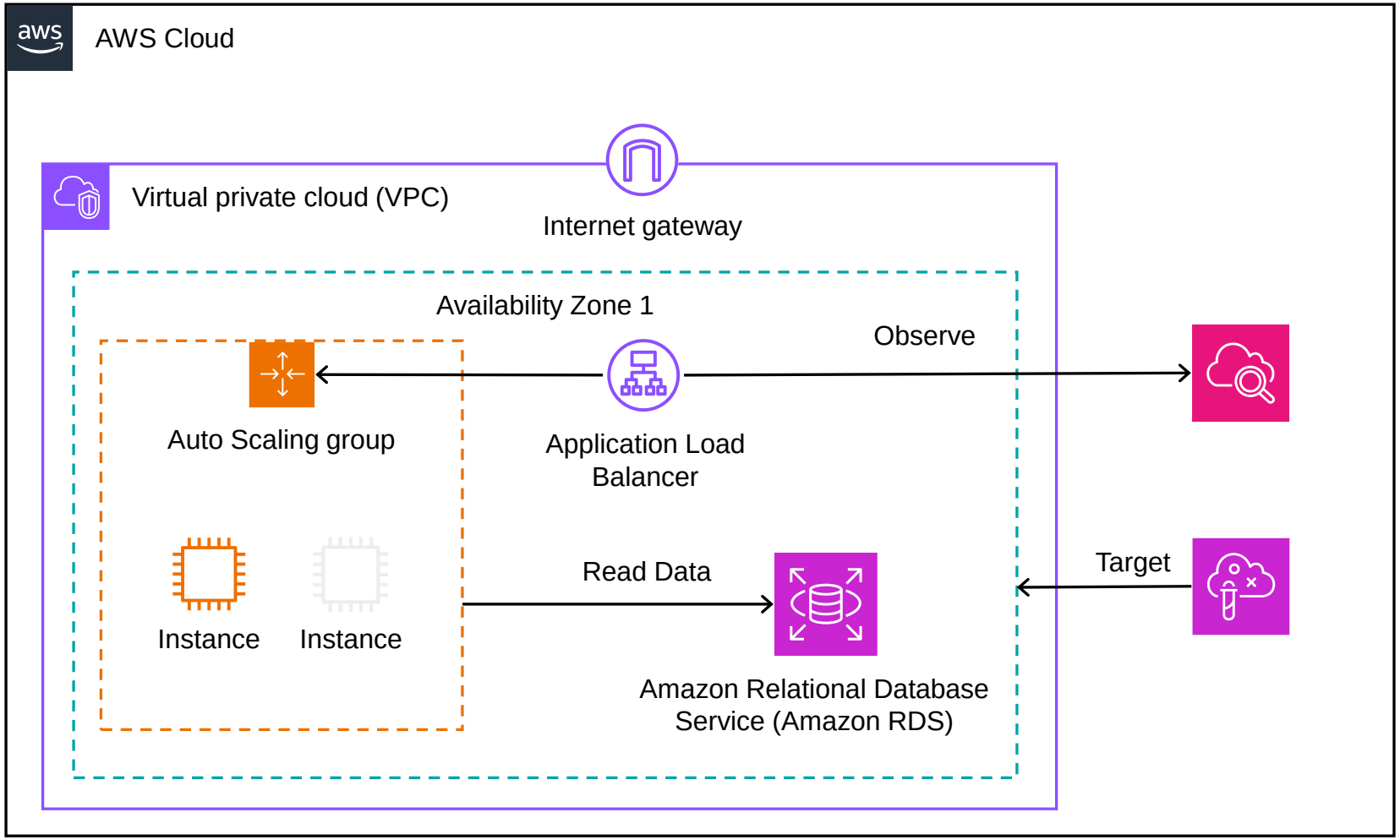
Experiment



ALB 입장에선,

아직 하나가 있으니

당연히 OK!

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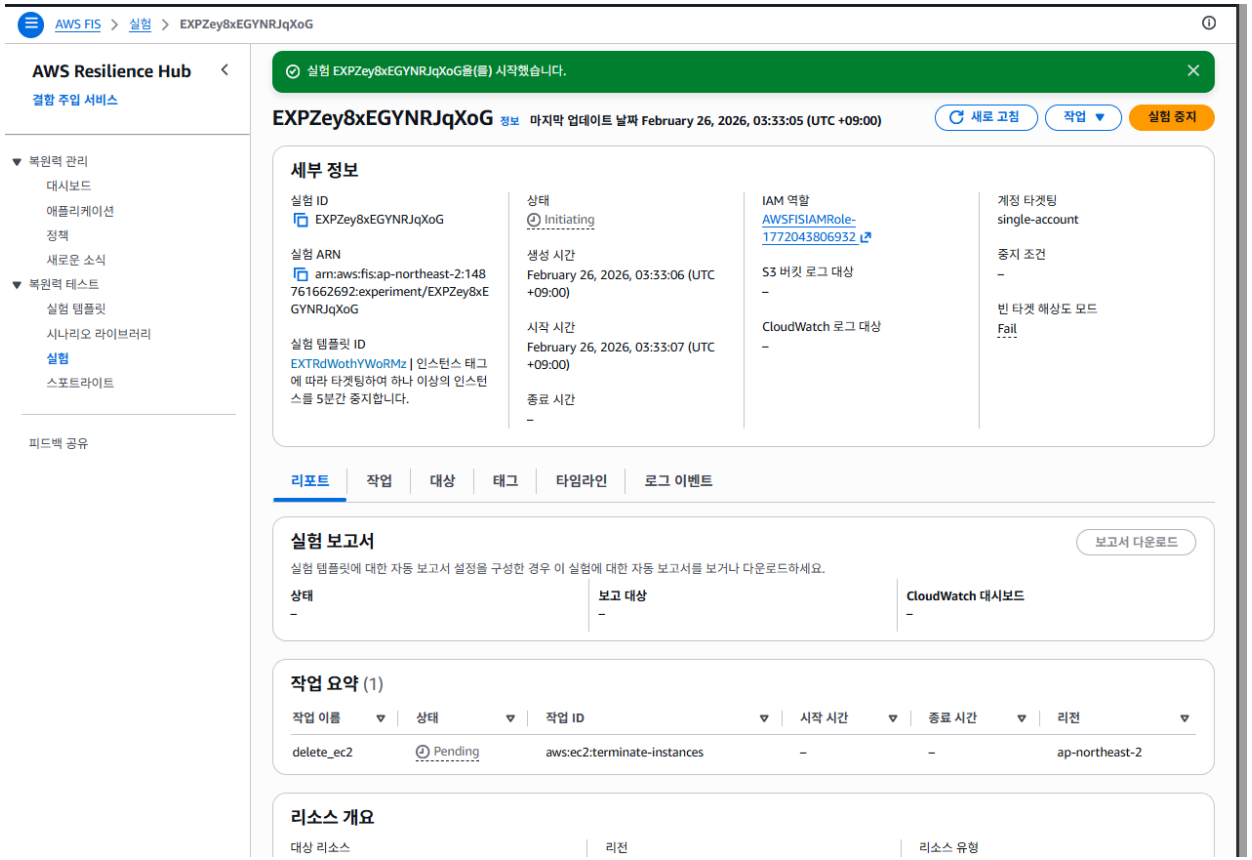
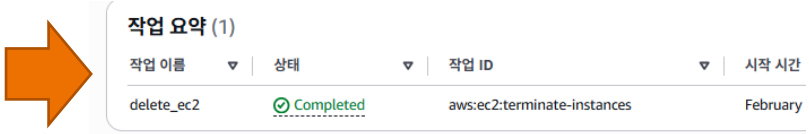
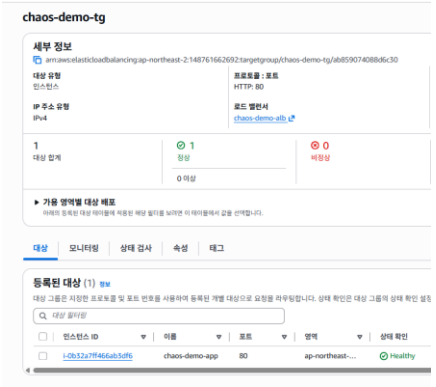
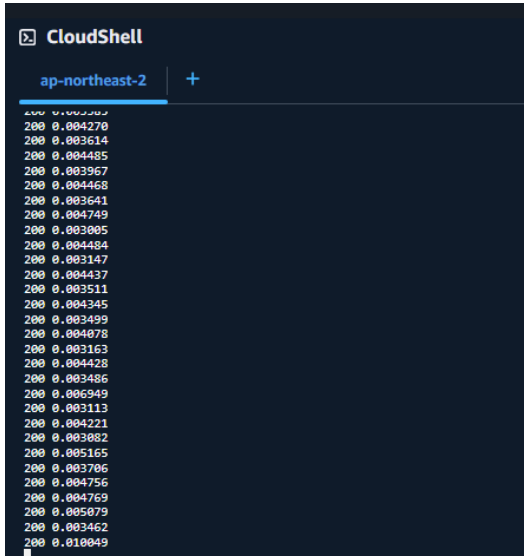
Experiment









하나의 인스턴스가 남아있어. 정상!

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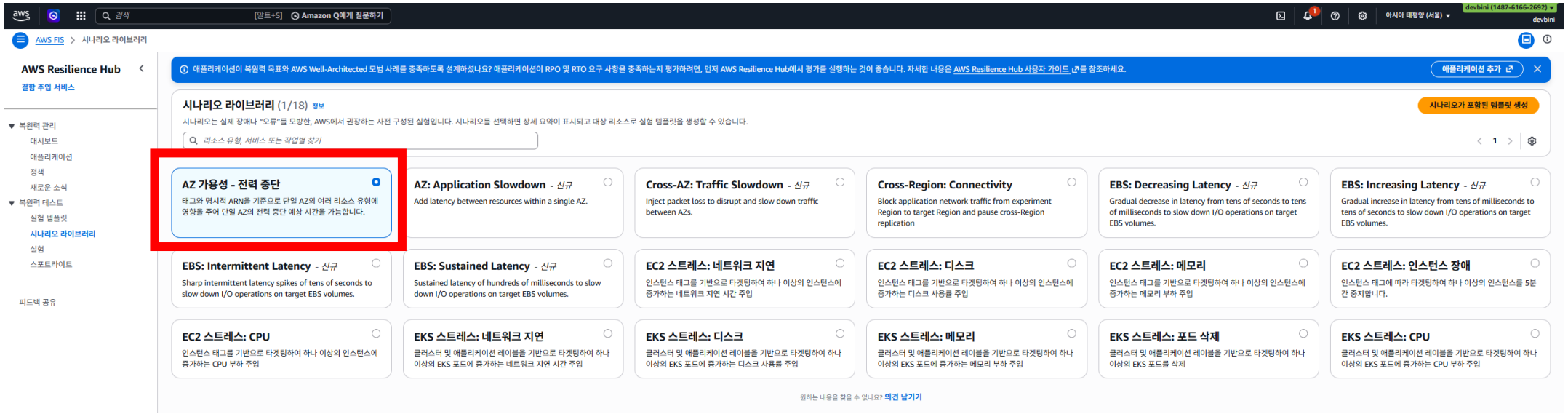
Experiment



이번엔, AZ를 부숴봅시다.

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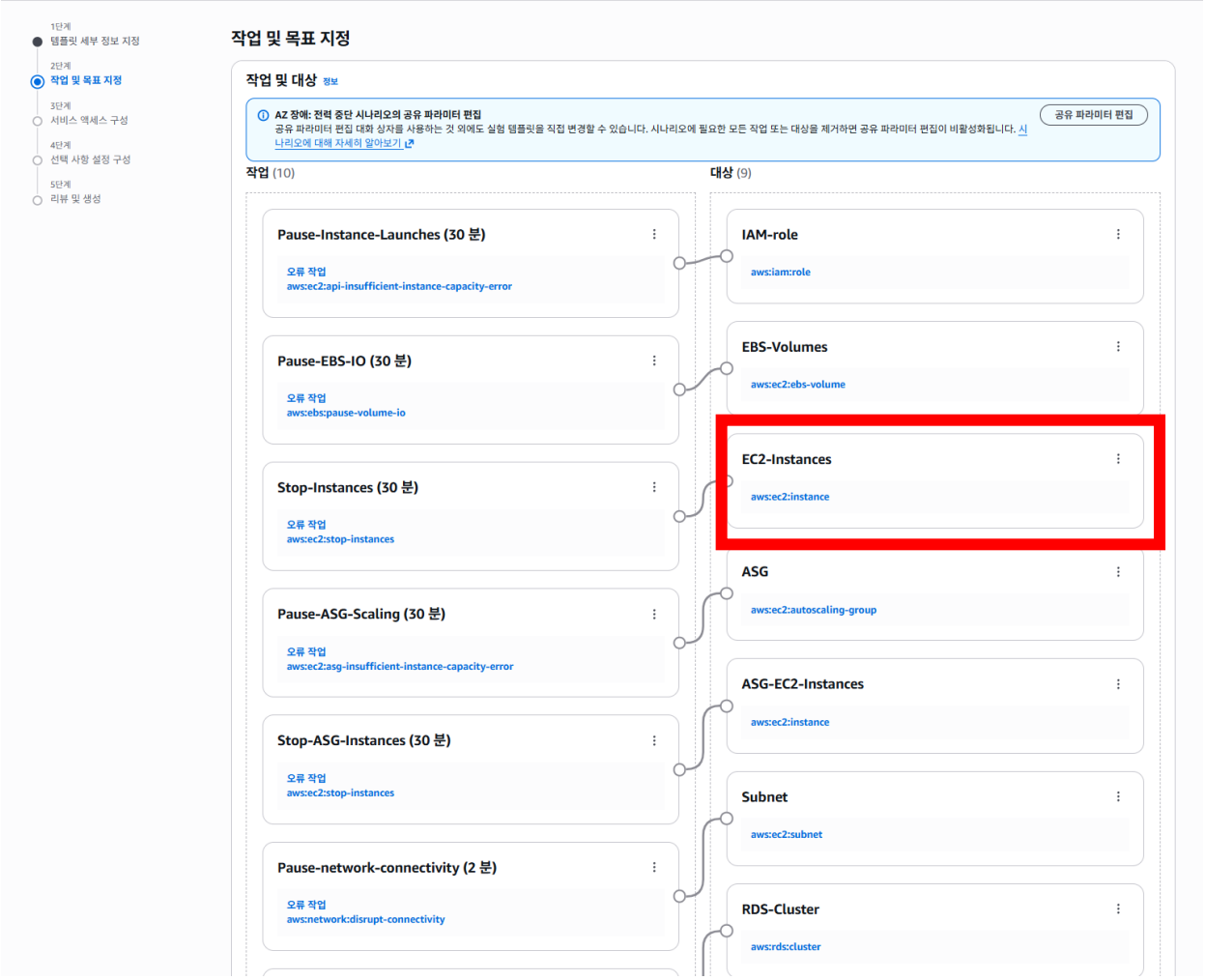
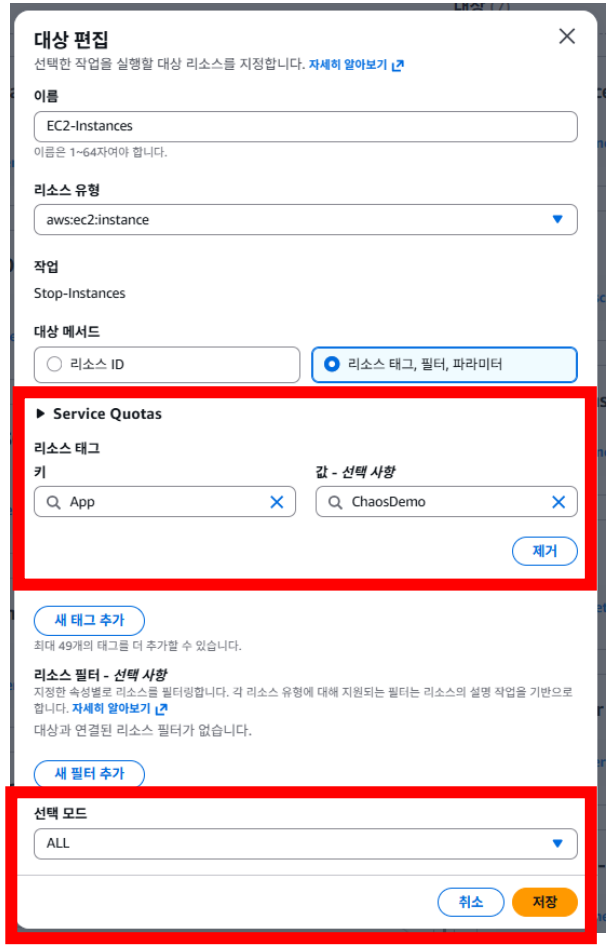
Experiment





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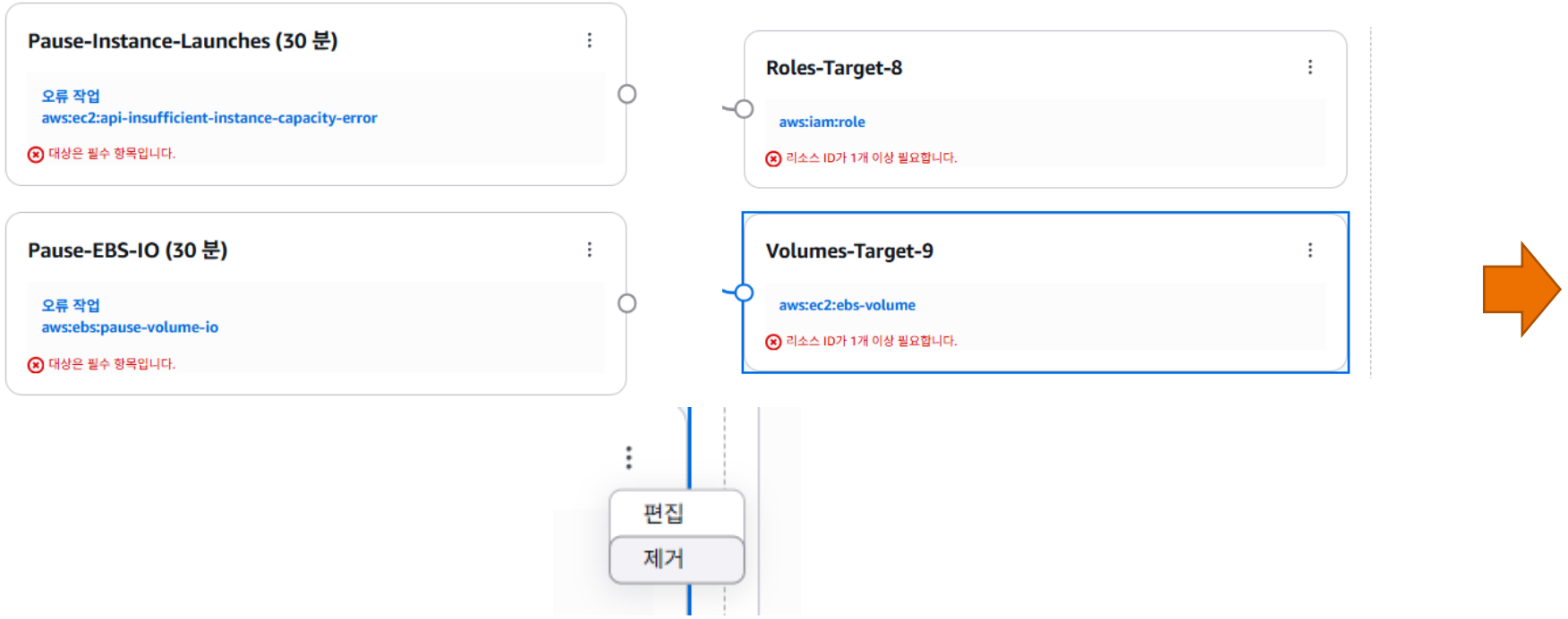
Experiment



테스트용이니, 위 요소들은 제거 해 주세요.

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Experiment

이번 장애는 시그널링이 사용하는 인프라(AWS)의 특정 리전 제어하기 어려운 **불가항력적인 장애**이지만, 그럼에도 불구하다.

어떻게 될까?

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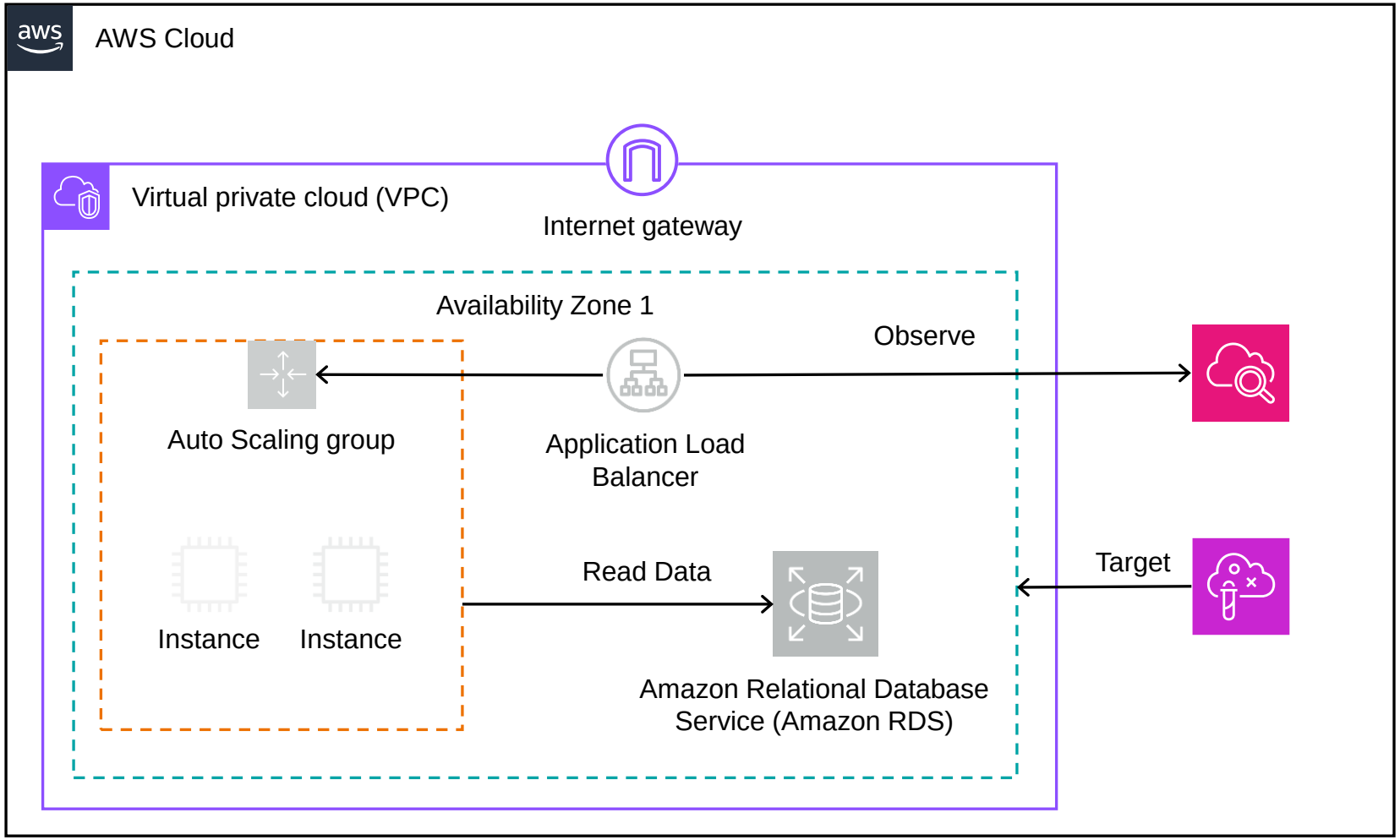
Experiment



AWS 입장에선

AZ-1에 있는 모든 리소스가

죽어버린 상황.

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Experiment

EXP3nRnVchdNkMLrSW 정보 마지막 업데이트 날짜 February 26, 2026, 04:03:02 (UTC +09:00)

새로 고침 | 작업 | 실험 중지

세부 정보

실험 ID	상태	IAM 역할	계정 타겟팅
EXP3nRnVchdNkMLrSW	Running	AWSFISIAMRole-1772045259501	single-account

실험 ARN
arn:aws:fis:ap-northeast-2:148761662692:experiment/EXP3nRnVchdNkMLrSW

생성 시간
February 26, 2026, 04:01:10 (UTC +09:00)

S3 버킷 로그 대상
–

중지 조건
–

실험 템플릿 ID
EXT29yYWPtx9Q2fA | 태그와 명시적 ARN을 기준으로 단일 AZ의 여러 리소스 유형에 영향을 주어 단일 AZ의 전력 중단 예상 시간을 가늠합니다.

시작 시간
February 26, 2026, 04:01:10 (UTC +09:00)

CloudWatch 로그 대상
–

빈 타겟 해상도 모드
Skip

종료 시간
–

작업 요약 (8)

작업 이름	상태	작업 ID	시작 시간	종료 시간	리전
FIS-Wait	Running	aws:fis:wait	February 26, 2026, 04:01:24 (UTC +09:00)	–	ap-northeast-2
Failover-RDS	Skipped	aws:rds:failover-db-cluster	–	–	ap-northeast-2
Pause-ASG-Scaling	Skipped	aws:ec2:asg-insufficient-instance-capacity-error	–	–	ap-northeast-2
Pause-ElastiCache	Skipped	aws:elasticache:replicationgroup-interrupt-az-power	–	–	ap-northeast-2
Pause-network-connectivity	Skipped	aws:network:disrupt-connectivity	–	–	ap-northeast-2
Stop-ASG-Instances	Skipped	aws:ec2:stop-instances	–	–	ap-northeast-2
Stop-Instances	Running	aws:ec2:stop-instances	February 26, 2026, 04:01:25 (UTC +09:00)	–	ap-northeast-2
Start-ARC-Zonal-Autoshift	Pending	aws:arc:start-zonal-autoshift	–	–	ap-northeast-2

```
200 0.002974
200 0.004050
200 0.002640
200 0.003587
200 0.002577
502 0.003841
502 0.002880
502 0.003731
502 0.002567
502 0.003715
502 0.003053
502 0.003668
502 0.002479
502 0.003889
```

실험 실행중인 결과…

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Experiment

에러는 얼마나 발생했는가?

복구까지 얼마나 오래 걸렸는가?

ASG가 재생성되는 데 의존성 문제는 없었는가?

헬스체크는 제대로 의도대로 동작했는가?

"결과분석"

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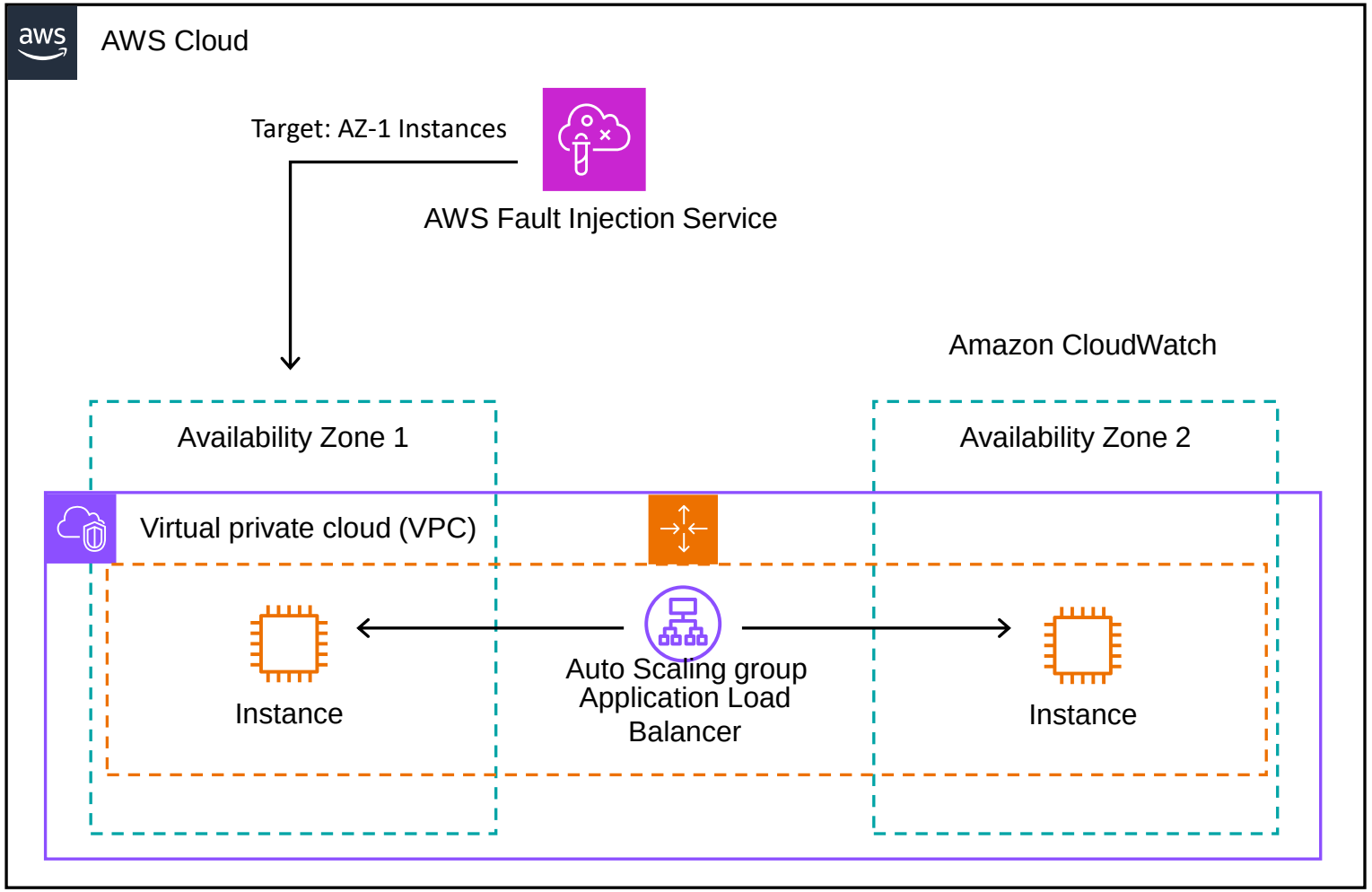
Experiment



"이렇게 구성하면 되겠구나~!"

-> 문제점 수정 근거 획득

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Experiment

AWS 지역을 사용할 수 없게 되는 경우는 극히 드물지만 어쨌든 발생하는 일

"지역 단위의 중단을 시뮬레이션하는 실험을 정기적으로 실행함으로써 조기에 모든 시스템 약점을 파악해 수정할 수 있었다. US-EAST-1이 실제로 사용할 수 없게 되는 상황이 발생했을 때 넷플릭스 시스템은 충분히 견고했기 때문에 트래픽 페일오버를 원활하게 처리할 수 있었다"

넷플릭스, 카오스 콩 (2015)

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So…

그래서…

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So.

Chaos Engineering은 엔지니어 측면에선 큰 기술이 아님.

우리는 장애를 피하는 팀인가?
아니면 장애를 견디는 팀인가?

결국 Chaos Engineering이란,
“괜찮겠지” 가 “확인했다” 가 되는 과정임.

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